

# Brent Pappas

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Portfolio: <https://www.pappasbrent.com>

GitHub: <https://www.github.com/PappasBrent>

## EDUCATION

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**PhD.** Computer Science. GPA: 3.983. University of Central Florida. Orlando, Florida. August 2021 - Present.

**Master of Science.** Computer Science. GPA: 3.983. University of Central Florida. Orlando, Florida. August 2021 - December 2025.

**Bachelor of Science.** Computer Science. GPA: 3.849. University of Central Florida. Orlando, Florida. August 2017 - May 2021.

## TEACHING

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| <b>Spring 2026</b>            | <b>Instructor of Record.</b> COP-3402 Systems Software. UCF. <a href="#">Course homepage</a> . <a href="#">Lecture Recordings</a> . Mean average teaching evaluation score: <b>4.62/5</b> . 164/250 enrolled students responded. Department mean was 4.24, college mean 4.21. <a href="#">Student ratings</a> . <a href="#">Student comments</a> . |
| <b>Fall 2025</b>              | <b>Graduate Teaching Assistant.</b> COP-3402 Systems Software. UCF. Mean average overall teaching evaluation score: <b>4.93/5</b> . 15 students responded. <a href="#">Student perception of instruction</a> .                                                                                                                                     |
| <b>Spring 2025</b>            | <b>Graduate Teaching Assistant.</b> COP-3402 Systems Software. UCF.                                                                                                                                                                                                                                                                                |
| <b>Spring 2021</b>            | <b>Peer Tutor Mentor.</b> COP-3502 Computer Science 1 & COP-3330 Object Oriented Programming. UCF.                                                                                                                                                                                                                                                 |
| <b>Spring &amp; Fall 2020</b> | <b>Peer Tutor.</b> COP-3502 Computer Science 1 & COP-3330 Object Oriented Programming. UCF.                                                                                                                                                                                                                                                        |
| <b>Fall 2019</b>              | <b>Peer Tutor.</b> COP-3502 Computer Science 1. UCF.                                                                                                                                                                                                                                                                                               |

## PUBLICATIONS

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- 2026 Build Code is Still Code: Finding the Antidote for Pipeline Poisoning:** by Brent Pappas and Paul Gazzillo. ICSE NIER. Acceptance rate of 33.11% (50 / 151). [Pre-print](#). [Artifact](#). Badges: Available and reusable. [Presentation](#). [Slides](#).
- 2024 Semantic Analysis of Macro Usage for Portability:** by Brent Pappas and Paul Gazzillo. ICSE. Acceptance rate of 22.26% (234 / 1051). [Pre-print](#). [Artifact](#). Badges: Available and reusable. [Presentation](#). [Slides](#).
- 2023 Holy Macroni! A recipe for progressive language enhancement:** by Brent Pappas. Blog post for Trail of Bits. [Blog post](#). [Artifact](#).
- 2022 SugarC: Scalable Desugaring of Real-World Preprocessor Usage into Pure C:** by Zach Patterson, Zenong Zhang, Brent Pappas, Shiyi Wei, and Paul Gazzillo. ICSE. Acceptance rate of 28.51% (197 / 691). [Pre-print](#). [Artifact](#). Badges: Available and reusable. [Presentation](#).

## SERVICE

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- March 2026** **NAC - NAPC Volunteer.** Orlando, Florida.
- **Inflating Balloons:** Inflated competition balloons for the 2026 North American Championship and Programming Camp (NAC-NAPC), hosted at UCF.
- October 2025** **UCF CS Grad Symposium 2025 Student Panelist.** Orlando, Florida.
- **Early PhD Advice:** Recommended that new graduate students remain humble and start honing their communication skills immediately after starting a PhD.
- August 2025** **UCF GradLaunch PhD Panelist.** Orlando, Florida.
- **Advising:** Instructed new graduate students how to manage their time and organize their day in order to be productive researchers.
- August 2025** **UCF "Surviving Graduate School" Panelist.** Orlando, Florida.
- **Community engagement:** Taught prospective graduate students how to become active in the UCF community.

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| June 2025      | <b>Camp Connect Lab Leader.</b> Orlando, Florida.<br><ul style="list-style-type: none"> <li>• <b>Human Computer Interaction Lessons:</b> Gave primary school students hands-on training with biometric equipment, including respiration belts, EDA and ECG sensors, fNIRS headbands, and eye-tracking sensors.</li> </ul> |
| May 2025       | <b>UCF Graduate School Info Session PhD Panelist.</b> Orlando, Florida.<br><ul style="list-style-type: none"> <li>• <b>Advising:</b> Answered undergraduate students' questions about UCF's Computer Science PhD program.</li> </ul>                                                                                      |
| April 2024     | <b>UCF CECS Senior Design Judge.</b> Orlando, Florida.<br><ul style="list-style-type: none"> <li>• <b>Judging:</b> Assessed capstone projects for seniors of the College of Engineering and Computer Science.</li> </ul>                                                                                                  |
| October 2022   | <b>UCF CECS Career Kickoff Alumni Mentor.</b> Virtual.<br><ul style="list-style-type: none"> <li>• <b>Mentoring:</b> Advised UCF freshmen on goal-setting, resume-building, and overall professionalism.</li> </ul>                                                                                                       |
| May 2024       | <b>NAC - NAPC Volunteer.</b> Orlando, Florida.<br><ul style="list-style-type: none"> <li>• <b>Competition setup:</b> Moved computers and cables to the competition room for the 2024 North American Championship and Programming Camp (NAC-NAPC), hosted at UCF.</li> </ul>                                               |
| 2024           | <b>OOPLSA 2024 Artifact Evaluation Committee Member.</b> Virtual.<br><ul style="list-style-type: none"> <li>• <b>Artifact Review:</b> Reviewed code associated with papers accepted to OOPLSA 2024. Distinguished reviewer.</li> </ul>                                                                                    |
| June 2023      | <b>Camp Connect Lab Leader.</b> Orlando, Florida.<br><ul style="list-style-type: none"> <li>• <b>Javascript lessons:</b> Taught primary school students the basics of computer programming in JavaScript.</li> </ul>                                                                                                      |
| Nov - Dec 2022 | <b>EuroSys 2023 Shadow Program Committee Member.</b> Virtual.<br><ul style="list-style-type: none"> <li>• <b>Paper review:</b> Reviewed submissions to EuroSys 2023.</li> </ul>                                                                                                                                           |
| June 2021      | <b>Camp Connect Lab Leader.</b> Orlando, Florida.<br><ul style="list-style-type: none"> <li>• <b>Computer programming lessons:</b> Engaged primary school students in computer programming lessons through a series of programming-themes games.</li> </ul>                                                               |

## PRESENTATIONS

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| 2026 | <b>Build Code is Still Code: Finding the Antidote for Pipeline Poisoning.</b> Presentation at the 2026 International Conference on Software Engineering in Rio de Janeiro, Brazil. <a href="#">YouTube recording</a> . <a href="#">Slides</a> . |
| 2026 | <b>Curing Pipeline Poisoning.</b> 3 minute thesis on my research on securing build systems against pipeline poisoning with build phase isolation. Presented at the CRA 2026 Grad Cohort Workshop. <a href="#">Recording link</a> .              |
| 2026 | <b>Invited Talk - The PhD Life.</b> Guest lecture on graduate student life presented to Sonoma State University's 2026 Spring Colloquium, hosted by Dr. John Sullins. <a href="#">Slides</a> .                                                  |
| 2025 | <b>Invited Talk - The PhD Life.</b> Guest lecture on graduate student life presented to Dr. Suzanne Rivoire's undergraduate course, Computing Professions, at Sonoma State University. <a href="#">Slides</a> .                                 |
| 2024 | <b>Semantic Analysis of Macro Usage for Portability (ICSE 2024).</b> Presentation at the 2024 International Conference on Software Engineering in Lisbon, Portugal. <a href="#">Recording</a> .                                                 |
| 2023 | <b>Semantic Analysis of Macro Usage for Portability (CAE-R)</b> Virtual presentation for Center of Academic Excellence in Cyber Research (CAE-R). <a href="#">Recording</a> ; <a href="#">jump to 25:38</a> .                                   |

## POSTERS

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| 2024 | <b>Holy Macroni! Finding Macro Usage Errors in the Linux Kernel.</b> 2024 International Symposium on Secure and Private Execution Environment Design. <a href="#">Link</a> . |
| 2022 | <b>Cpp2C: Transforming C Preprocessor Macros to C Language Constructs.</b> 2022 University of Central Florida Student Scholar Symposium. <a href="#">Link</a> .              |

## HONORS AND AWARDS

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| 2026 | <b>GTA Excellence Award (University level).</b> For providing exceptional teaching assistance to the course Systems Software. Awarded to 1 out of 800+ graduate teaching assistants. <a href="#">Award letter</a> . |
| 2026 | <b>GTA Excellence Award (College level).</b> For providing exceptional teaching assistance to the course Systems Software. <a href="#">Award letter</a> .                                                           |
| 2026 | <b>CRA-WP 2026 Grad Cohort Workshop Invitee.</b> The CRA-WP arranged and paid for me to travel to the 2026 CRA-WP Grad Cohort Workshop in Seattle, Washington, because of my superior workshop application.         |

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| <b>2024</b>        | <b>UCF 3MT Finalist.</b> For effectively communicating my research to a general audience in a 3 minute talk. <a href="#">Video Certificate</a> .                        |
| <b>2024</b>        | <b>Distinguished Artifact Reviewer.</b> For my attention to detail while reviewing artifacts for OOPLSA 2024. <a href="#">Link</a> .                                    |
| <b>2024</b>        | <b>FCI Student Scholarship.</b> For my interest in interdisciplinary research, scholarship, and collaboration. \$5000 award.                                            |
| <b>2021</b>        | <b>Cum laude honors.</b> For having a cumulative undergraduate GPA greater than 3.8.                                                                                    |
| <b>2017 - 2021</b> | <b>Deans' list.</b> For continued undergraduate scholastic achievement.                                                                                                 |
| <b>2017</b>        | <b>Burnett Honors Scholar.</b> For academic talent, motivation, intellectual curiosity and creativity.                                                                  |
| <b>2017</b>        | <b>Provost Scholarship.</b> For outstanding academic performance in high school. Completely covered my undergraduate tuition fees at the University of Central Florida. |

## CERTIFICATIONS

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[Completed higher education training, Preparing Tomorrow's Faculty](#)

[Level 1 College Readiness and Learning Assessment \(CRLA\) certified tutor.](#)

[Level 2 College Readiness and Learning Assessment \(CRLA\) certified tutor.](#)

UCF Student Enhancement Training (SET) program certified.

## WORKSHOPS

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|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
| <b>March 2026</b>     | 2026 CRA-WP Grad Cohort Workshop. Seattle, Washington.                 |
| <b>Oct - Nov 2025</b> | 2025 CRA-WP Virtual Mentoring Series Workshops. Virtual.               |
| <b>May 2025</b>       | 2025 CRA-E Career Landscape Workshop. Virtual.                         |
| <b>June 2022</b>      | Programming Language Mentoring Workshop (PLMW). San Diego, California. |

## COMPLETE WORK EXPERIENCE

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| <b>Jan 2026 - Present</b>  | <b>Graduate Teaching Associate (Instructor of Record) for COP-3402 Systems Software</b><br>University of Central Florida. Orlando, Florida. |
| <b>Aug 2024 - May 2025</b> | <b>Graduate Research Assistant</b><br>University of Central Florida. Orlando, Florida.                                                      |
| <b>Aug 2023 - Apr 2024</b> | <b>Graduate Teaching Assistant for COP-3402 Systems Software</b><br>University of Central Florida. Orlando, Florida.                        |
| <b>May 2023 - Aug 2023</b> | <b>Software Engineering Intern</b><br>Trail of Bits. Remote.                                                                                |
| <b>Jul 2021 - May 2023</b> | <b>Graduate Research Assistant</b><br>University of Central Florida. Orlando, Florida.                                                      |
| <b>Sep 2019 - Jul 2021</b> | <b>Quality Assurance Coordinator</b><br>SP+. Orlando, Florida.                                                                              |
| <b>Jan 2021 - May 2021</b> | <b>Peer Tutor Mentor</b><br>Student Academic Resource Center at the University of Central Florida. Orlando, Florida.                        |
| <b>Aug 2018 - May 2021</b> | <b>Peer Tutor</b><br>Student Academic Resource Center at the University of Central Florida. Orlando, Florida.                               |

## ADVISING

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### UNDERGRADUATE RESEARCHERS

**Aaron Burgess.** 2026 - Present. Human factors in software engineering.

**Thiago Airoidi.** 2026 - Present. Analysis and security of software development pipelines.

**Rasmin Ahmed.** 2026 - Present. Analysis and security of software development pipelines.

**Daniel Henriquez.** 2026 - Present. Analysis and security of software development pipelines.

**Brayden Coggin.** 2026 - Present. Analysis and security of software development pipelines.

**Thai Nguyen.** 2025 - 2026. Human factors in software engineering and analysis and security of software development pipelines.

**Nikola Krstajic.** 2025. Human factors in software engineering.

**Michael Johnson.** 2025 - Present. Human factors in software engineering and analysis and security of software development pipelines.

**Layla Le.** 2025. Human factors in software engineering.

**Kenneth Valladares.** 2025. Human factors in software engineering.

**Julian Velez.** 2025. Human factors in software engineering.

**Jorge Toledo.** 2025. Human factors in software engineering.

**Hasini Daliboyina.** 2025. Human factors in software engineering.

**David Gusmao.** 2025 - Present. Human factors in software engineering and analysis and security of software development pipelines.

**Anju Thomas.** 2025. Human factors in software engineering.

**Adam Betinsky.** 2025 - Present. Human factors in software engineering and analysis and security of software development pipelines.

**Zachary Burkett.** 2024. Semantics-aware translation of C preprocessor.

**Joseph Zalusky.** 2024. Semantics-aware translation of C preprocessor macros.

## PROJECTS

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**2026 Foreman.** University of Central Florida. A Python tool that enables developer to specify the commands file access permissions of phase of a project's build system, to protect these phases against pipeline poisoning attacks. Foreman is the tool associated with 2026 ICSE NIER paper, *Build Code is Still Code: Finding the Antidote for Pipeline Poisoning*.

**2024 Macroni.** Trail of Bits. A C static analysis tool that combines Trail of Bits' tools PASTA and VAST to lower C preprocessor macros down to MLIR. [GitHub](#).

**2024 Maki.** University of Central Florida. A C++ Clang plugin that analyzes hows C preprocessor macros affect the C AST. Maki provides language porting tools information they can use to port macros to target languages in a way that preserves not just macro behavior, but also macro abstractions. Maki is the tool associated with 2024 ICSE paper, *Semantic Analysis of Macro Usage for Portability*, for which I am the lead-author. [GitHub](#).

## SKILLS SUMMARY

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**Languages:** Python3, C/C++, Bash, and HTML/CSS/JavaScript

**Tools:** Git, GitHub, and LLVM/Clang

**Platforms:** Linux, Web, and Windows

**Soft Skills:** Leadership, Writing, Communication, and Time management